

SCOPE Retrofit Measure Analysis Report

Comprehensive Energy and Financial Analysis

Executive Summary

This report compares energy consumption and operational costs between baseline and all applied renovation scenarios from CSV results. Embodied carbon is annualized using an analysis period of 30 years, while operational carbon is reported over 1 year. The construction cost data is from 2026 RSMeans Database. The embodied carbon is calculated using data from EC3 Environmental Product Declaration (EPD) Database. Since RSMeans Database is proprietary, users can adopt customized cost dataset when needed.

Building Information

- Weather File: USA_TX_Amarillo-Husband.Amarillo.Intl.AP.723630_TMYx.epw
- Building Name: SmallOffice
- Building Location: Amarillo, TX
- Total Floor Area: 511 m²

Renovation Details by Scenario

Scenario	Renovation Type	Renovation Details
Baseline	baseline	Baseline (no envelope renovation)
Scenario 1	wall + roof	Wall upgrade: Exterior walls were upgraded to R-10.0 with Fiberglass Batts insulation. Roof upgrade: The roof was upgraded to R-20.0 using Blown Fiberglass insulation.
Scenario 2	wall + roof	Wall upgrade: Exterior walls were upgraded to R-13.0 with Fiberglass Batts insulation.

		Roof upgrade: The roof was upgraded to R-24.4 using Blown Fiberglass insulation.
Scenario 3	wall + roof	Wall upgrade: Exterior walls were upgraded to R-13.0 with Expanded Polystyrene (EPS) Foam Board insulation. Roof upgrade: The roof was upgraded to R-24.4 using Extruded Polystyrene (XPS) Foam Board insulation.
Scenario 4	wall + roof	Wall upgrade: Exterior walls were upgraded to R-20.0 with Fiberglass Batts insulation. Roof upgrade: The roof was upgraded to R-30.0 using Blown Fiberglass insulation.
Scenario 5	wall + roof + window + door	Wall upgrade: Exterior walls were upgraded to R-13.0 with Fiberglass Batts insulation. Roof upgrade: The roof was upgraded to R-24.4 using Blown Fiberglass insulation. Window upgrade: - Upgraded to 3-pane glazing - Weatherstripping: silicone adhesive smoke gasket - Glazing film: low-e film - Window frame: wood window frame - Caulking: acrylic - Infiltration reduction: 30% Door upgrade: - Door type: wooden door - Bottom seal: automatic door bottom - Top/side seals: jamb weatherstrip - Infiltration reduction: 30%

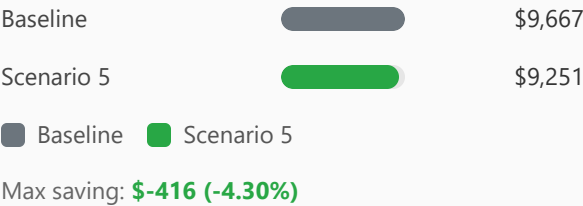
Annual Energy Analysis

Scenario	Metric	Baseline	Renovation	Delta	Savings %
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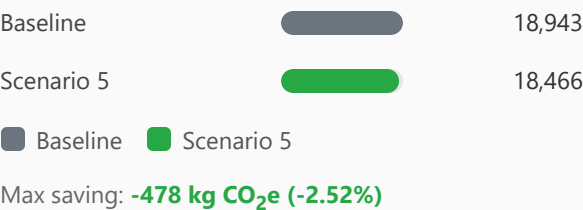
Scenario 1	Total Site Energy (GJ)	195.48	193.42	2.06	1.05%
Scenario 2	Total Site Energy (GJ)	195.48	192.49	2.99	1.53%
Scenario 3	Total Site Energy (GJ)	195.48	192.49	2.99	1.53%
Scenario 4	Total Site Energy (GJ)	195.48	191.34	4.14	2.12%
Scenario 5	Total Site Energy (GJ)	195.48	190.41	5.07	2.59%

Key Performance Metrics

Operational Cost Saving (Scenario - Baseline) (\$/yr) Comparison



Operational Carbon Saving (Scenario - Baseline) (kg CO₂e/yr) Comparison



Min Retrofit Construction Cost

\$45,576

Scenario 1

Min Retrofit Embodied Carbon

5,558 kg CO₂e

Embodied carbon intensity (ECI): 11 kg CO₂e/m²
Scenario 1

Lowest Retrofit Cost Payback Period

213 years

Scenario 4

Lowest Retrofit Carbon Payback Period

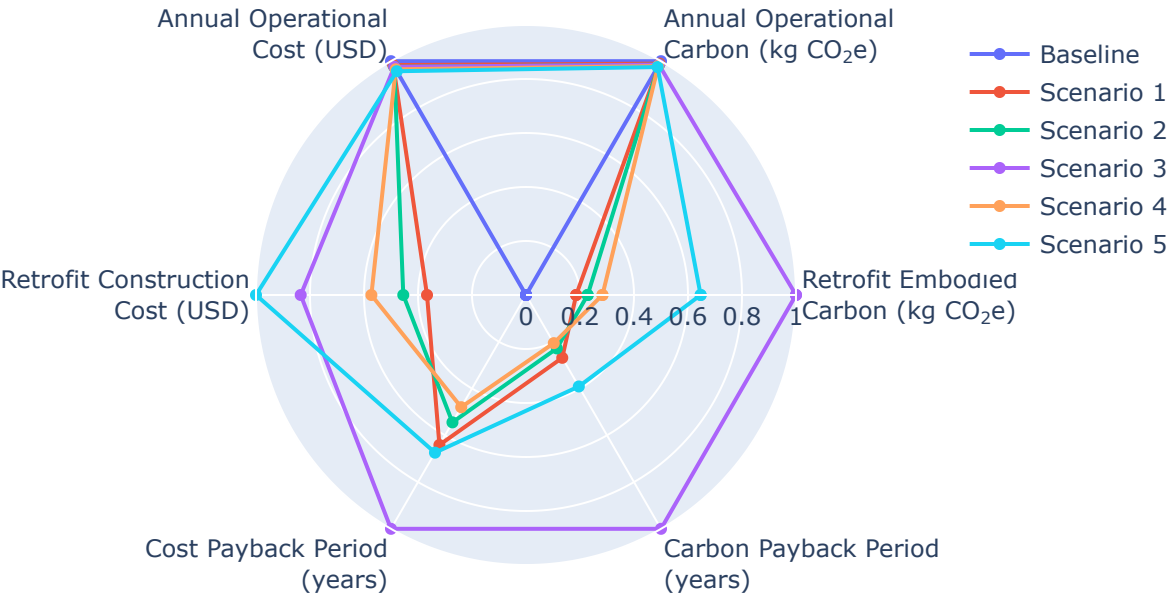
21 years

Scenario 4

Comparative Visualizations between Renovation Scenarios

Spider Chart Visualization

Parametric Scenario Comparison (Normalized Spider Chart)



Payback Period of the Retrofit

Cost Payback Period

Payback = retrofit construction cost / abs(Operational Cost Saving (Scenario - Baseline) (\$/yr)).

Scenario 1		284 yrs
Scenario 2		241 yrs
Scenario 3		443 yrs
Scenario 4		213 yrs
Scenario 5		299 yrs

Carbon Payback Period

Payback = retrofit embodied carbon / abs(Operational Carbon Saving (Scenario - Baseline) (kg CO₂e/yr)).

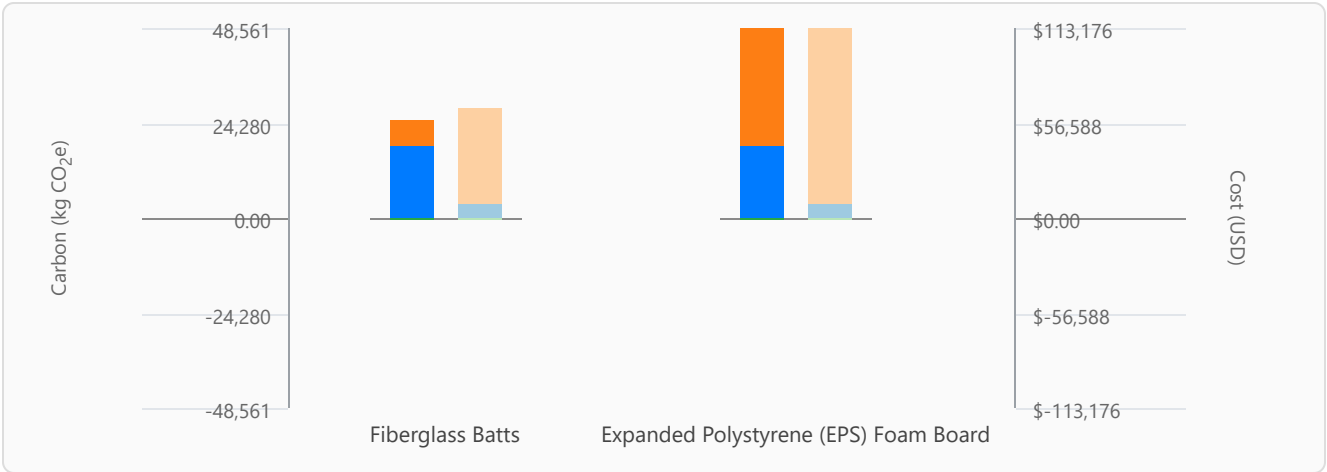
Scenario 1		28 yrs
Scenario 2		24 yrs
Scenario 3		104 yrs
Scenario 4		21 yrs
Scenario 5		41 yrs

Material Type Comparison

Auto-generated when scenarios share the same target insulation settings but use different insulation materials.

Wall insulation material comparison (target R-13.0)

Scenario	Insulation Material Type	Operational Cost Saving (Scenario - Baseline) (\$/yr)	Operational Carbon Saving (Scenario - Baseline) (kg CO ₂ e/yr)	Cost Payback Period (years)	Carbon Payback Period (years)
Scenario 2	Fiberglass Batts	\$-234	-289	241 yrs	24 yrs
Scenario 3	Expanded Polystyrene (EPS) Foam Board	\$-234	-289	443 yrs	104 yrs



- Annual operational carbon (positive)
- Retrofit embodied carbon (positive)
- Annual operational carbon saving (negative)
- Annual operational cost (positive)
- Retrofit construction cost (positive)
- Annual operational cost saving (negative)

Material List

Material properties and consumption by renovation scenario. Missing values are shown as N/A.

Scenario	Component	Material	Volume (m ³)	Area (m ²)	Thickness (m)	Length (m)	Thermal Conductivity (W/m·K)	Density (kg/m ³)
Scenario 1	Wall insulation	Fiberglass Batts	N/A	282	N/A	N/A	N/A	N/A

Scenario 1	Roof insulation	Blown Fiberglass	N/A	599	N/A	N/A	N/A	N/A
Scenario 2	Wall insulation	Fiberglass Batts	N/A	282	N/A	N/A	N/A	N/A
Scenario 2	Roof insulation	Blown Fiberglass	N/A	599	N/A	N/A	N/A	N/A
Scenario 3	Wall insulation	Expanded Polystyrene (EPS) Foam Board	N/A	282	N/A	N/A	N/A	N/A
Scenario 3	Roof insulation	Extruded Polystyrene (XPS) Foam Board	N/A	599	N/A	N/A	N/A	N/A
Scenario 4	Wall insulation	Fiberglass Batts	N/A	282	N/A	N/A	N/A	N/A
Scenario 4	Roof insulation	Blown Fiberglass	N/A	599	N/A	N/A	N/A	N/A
Scenario 5	Wall insulation	Fiberglass Batts	N/A	282	N/A	N/A	N/A	N/A
Scenario 5	Roof insulation	Blown Fiberglass	N/A	599	N/A	N/A	N/A	N/A
Scenario 5	Window weatherstripping	silicone adhesive smoke gasket	N/A	N/A	N/A	N/A	N/A	N/A
Scenario 5	Window caulking	acrylic	N/A	N/A	N/A	N/A	N/A	N/A
Scenario 5	Door bottom seal	automatic door bottom	N/A	N/A	N/A	3.66	N/A	N/A
Scenario 5	Door top/side seal	jamb weatherstripping	N/A	N/A	N/A	16	N/A	N/A

Result Summary

Scenario	Retrofit Embodied Carbon (kg CO ₂ e)	Retrofit Construction Cost (USD)	Annual Operational Carbon (kg CO ₂ e)	Annual Operational Cost (USD)
Baseline	0.00	\$0.00	18,943	\$9,667
Scenario 1	5,558	\$45,576	18,743	\$9,507
Scenario 2	6,829	\$56,511	18,654	\$9,433
Scenario 3	29,907	\$103,743	18,654	\$9,433
Scenario 4	8,470	\$71,196	18,546	\$9,333
Scenario 5	19,345	\$124,324	18,466	\$9,251

Generated: April 09, 2026 | Report Type: Retrofit Impact Analysis | Run: run_test_003